WiFi Lab Questions Packet

*****DO NOT WRITE ON THIS PACKET*****

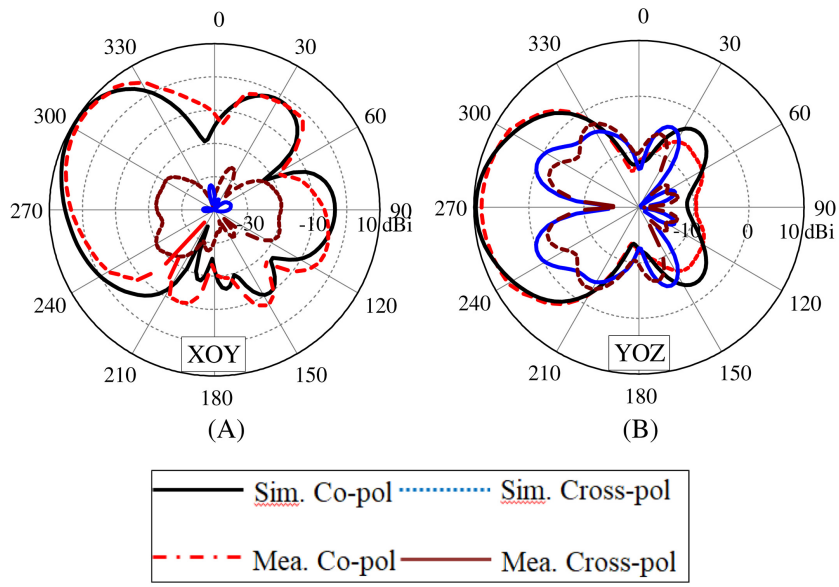
This questions packet contains 7 pages (including this cover page) and 11 questions. The total number of points on this exam is 70.

1. Sirius A, part of the Canis Major constellation, is the brightest star in the night sky. It's part of a binary system with a white dwarf, Sirius B. The spectrum of light from Sirius B, peaks at a wavelength of 116 nm. The star system is located $8.24 * 10^{13}$ km away from Earth.
   (a) (1 point) What is the frequency of the peak wavelength of light emitted from Sirius B?
   (b) (1 point) What is the energy of this wavelength of light?
   (c) (1 point) How long does it take for light emitted from the Sirius system to reach Earth?
   (d) (1 point) What region of the electromagnetic spectrum is this light in?
   (e) (1 point) Does this light have the same, less, or more energy than a radio wave?

2. A common antenna used in radio communication is the half-wave dipole antenna. Imagine you want to build a half-wave dipole antenna in order to transmit at 14.250 MHz.
   (a) (1 point) For a half-wave dipole transmitting at 14.250 MHz, what is the ideal length of the antenna?
   (b) (1 point) What is the relationship of the orientation of the magnetic field with the orientation of the dipole antenna?
   (c) (1 point) What is the real impedance for this ideal half-wave dipole antenna?
   (d) (1 point) If the antenna has an efficiency of $\epsilon_R = 0.83$ and transmits at 2.0 W, at what rate are photons transmitted from the antenna?
   (e) (1 point) What is the loss in dB of the antenna with an efficiency of $\epsilon_R = 0.83$?

3. Comparing the relative strength of antennas is an important aspect of antenna design. Two ways to measure the relative amount of energy radiated by an antenna are dBd and dBi.
   (a) (1 point) What is the difference between dBi and dBd?
   (b) (1 point) For an ideal dipole antenna, what is the dBd of the antenna in the direction of strongest propogation?
   (c) (1 point) Imagine a directional antenna with a dBi of 29. What is the dBd of this directional antenna?
   (d) (1 point) If you're feeding the 29 dBi directional antenna with 100 watts of power, what is the effected radiated power output compared to a dipole antenna?

4. The following diagram shows the radiation pattern for a four element microstrip Yagi antenna, a type of antenna that can be easily manufactured using a printed circuit board. The azimuthal plane of this antenna is the XY plane.



(a) (1 point) Is this antenna directional or omni-directional?

(b) (1 point) How many feedpoints does this variety of antenna require?

(c) (1 point) If this antenna were placed very close to the ground, how would the elevation radiation pattern of this antenna be affected?

(d) (1 point) What is the maximum gain of this antenna in dBi?

(e) (1 point) Would the maximum gain of this antenna increase or decrease if another director were added to this Yagi antenna?

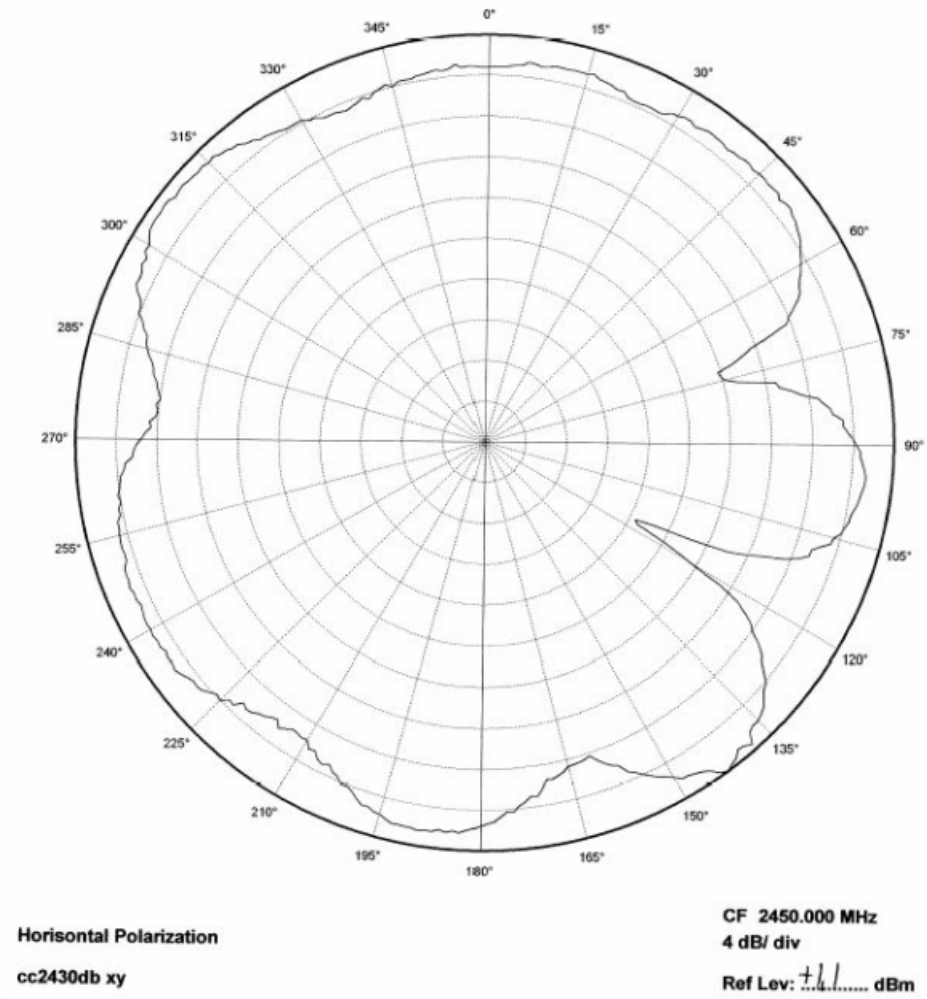
(f) (1 point) What is the azimuth of the direction of peak radiation of the antenna?

(g) (1 point) What is the half-power bandwidth of the antenna?

(h) (1 point) What is the front to back ratio of this antenna?

(i) (1 point) Why does the cross-pol pattern have a much smaller dBi than the co-pol pattern?

5. A common variety of WiFi antenna is the inverted F antenna. One particular inverted F antenna is sold by Texas Instruments and has the following radiation pattern.



(a) (1 point) In what direction is the direction of minimum radiation of the antenna?

(b) (1 point) What is the power in mW transmitted in the 270° direction?

(c) (1 point) What is the ratio of the power radiated by this antenna and an isotropic antenna assuming no losses in the antenna configuration?

(d) (1 point) Why might someone choose to use this antenna as a WiFi antenna over the Yagi antenna in the previous question?

6. Electromagnetic wave propagation is another important part of understanding antennas.
   (a) (1 point) What is the phase angle between the magnetic and electric fields of a photon?
   (b) (1 point) Write a relationship describing the speed of light in terms of other fundamental constants.
   (c) (1 point) In order to allow long distance radio wave communication, radio waves are reflected off the atmosphere. Which part of the atmosphere is most reflective?
   (d) (1 point) Compare the speed of propagation of radio waves in space and in the atmosphere.

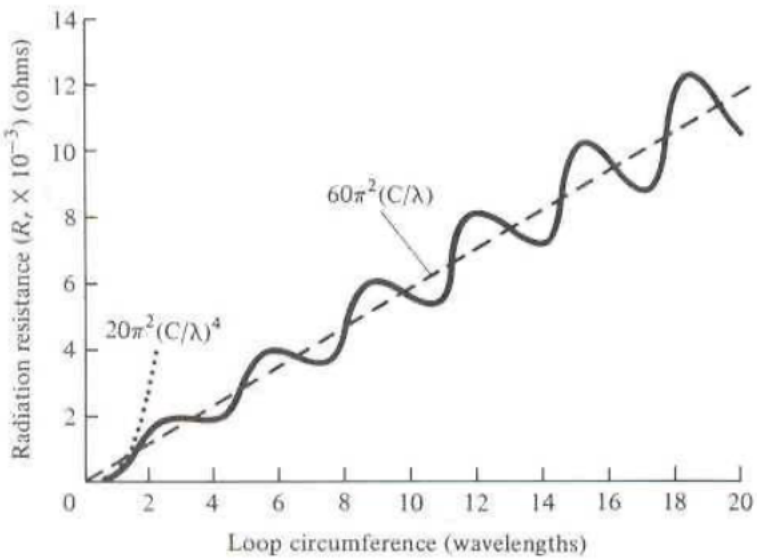
7. One of the most common types of antennas used for WiFi communication is the helical antenna. Imagine you are designing a simple helical antenna for use at the 5160 MHz WiFi channel.
   (a) (1 point) If we're going to operate the helical antenna in axial mode, which direction is the main propogation direction of the antenna?
   (b) (1 point) To properly operate in the axial mode, what is a possible circumference for the axial antenna?
   (c) (1 point) What is an appropriate pitch angle for such an antenna?
   (d) (1 point) For a pitch angle of 35°, 10 turns and a circumference of 11.0 cm, what length of wire is required to build this antenna?
   (e) (1 point) These helical antennas usually come with a ground plane. What is the purpose of a ground plane?
   (f) (1 point) What should be the minimum diameter of this ground plane?
   (g) (1 point) For the antenna specified in part d, what is an estimate for the directivity of the antenna?
   (h) (1 point) What is an estimate for the half power beamwidth?
   (i) (2 points) Sketch a potential radiation pattern for an antenna with these specifications. Make sure to indicate the orientation of the antenna with respect to the chart and use a logarithmic scale.

8. Another important part of antenna design is properly using antennas for different purposes. For the following questions, select which of the following antennas would be most appropriate for the application: helical, dipole, log-periodic, yagi, monopole.
   (a) (1 point) An antenna for far-range television reception across a large range of frequencies.
   (b) (1 point) An extremely cheap antenna used on a home WiFi router.
   (c) (1 point) An antenna used to communicate with the FalconSAT-3 satellite.
   (d) (1 point) An antenna used for general radio reception.
   (e) (1 point) An antenna used for WiFi communication with a stationary WiFi access point from a far distance.

9. Since antennas rely on alternating current to operate, impedance is an important value when considering antenna designs.
   (a) (1 point) For an electrical circuit with a resistance of 20 Ω and a reactance of 10 Ω, what is the magnitude of the impedance?
   (b) (1 point) What is the phase difference between the voltage and the current in the circuit?
   (c) (1 point) Is the circuit capacitive or inductive?
   (d) (1 point) What does SWR stand for?
   (e) (1 point) Imagine an antenna has an feedpoint impedence of 55 Ω and is fed with a lossless coaxial cable with an impedence of 50 Ω. What is the SWR of the system?
   (f) (1 point) For the above situation, what percentage of the power is reflected?
   (g) (1 point) If the antenna is connected to a 100W transmitter, how much power is radiated by the above antenna?

10. Rank these antennas from lowest directivity to highest directivity: dipole, dish, isotropic, yagi.
    (a) (4 points) Rank them.

11. Another very simple antenna is the circular loop antenna, which radiates by oscillating charge in an electrically conductive loop. For the purposes of this analysis, we're assuming the loop antenna wire is very thin and has a small diameter.
    (a) (1 point) Let ϕ be the angle describing the angular position in the loop antenna. We can assume the current in the antenna is constant with ϕ. What is the strength of the magnetic field parallel to radials of ϕ?
    (b) (1 point) What is the strength of the electric field parallel to the axis of the loop antenna?
    (c) (1 point) The two regions of interest in this antenna are the far field and near field. If the antenna transmits at 20m and the loop has a radius of 0.25m, what is the lower limit of the near field region?
    (d) (1 point) What is the upper limit of the near field region?
    (e) (1 point) What is the lower limit of the far field region?
    (f) (1 point) What is the upper limit of the far field region?



(g) (1 point) The radiation resistance of a loop antenna is shown above where the actual radiation resistance is shown by the solid black curve. What is the radiation resistance of a 10 wavelength circumference loop antenna?

(h) (1 point) The radiation resistance of a loop antenna can be approximated using a large loop approximation or a small loop approximation. Which approximation does the dashed line use?

(i) (1 point) For the antenna we specified above (20m wavelength and 0.25m radius), which approximation would be more appropriate?

(j) (1 point) The radiation resistance of the loop antenna is rather low. Without changing the size of the antenna or wavelength, how can the radiation resistance be increased?

(k) (1 point) The above analysis is based on the assumption of constant current within the loop antenna. However, this assumption is not accurate for larger loops. Why is this?

(l) (1 point) Would a larger loop have a higher or lower efficiency than a small loop?

(m) (1 point) Explain why a small loop antenna might be useful for triangulating the position of a radio signal.